



Research Paper

Numerical study of methane dry reforming in an electromagnetic induction heating reactor

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ABSTRACT

Efficient thermal management is critical for the dry reforming of methane (DRM), a highly endothermic process that converts greenhouse gases CH₄ and CO₂ into valuable syngas (H₂ and CO). Traditional heating methods often suffer from low energy efficiency and uneven temperature distribution, which hinder reactor performance. This study investigates electromagnetic induction heating as a high-performance alternative, offering rapid, non-contact, and localized energy delivery. A multiphysics model integrating electromagnetics, fluid dynamics, and chemical kinetics was developed to simulate an induction-heated DRM reactor. The model confirms effective energy conversion through the “electricity–magnetism–electricity–heat” pathway. Compared to conventional reactors, the induction-heated system achieved higher reaction zone temperatures and improved CH₄ and CO₂ conversion rates (10.81 % and 14.84 %, respectively) at an inlet temperature of 1000 K. Notably, this performance surpassed that of a conventional reactor operating at 1100 K. These results underscore the advantages of induction heating in enhancing syngas yield, thermal efficiency, and temperature uniformity, offering a promising approach for advancing DRM technology and sustainable greenhouse gas utilization.

1. Introduction

Dry reforming of methane (DRM), a key process for converting CH₄ and CO₂ into syngas (CO + H₂), has attracted significant attention owing to its dual benefits of greenhouse gas mitigation and sustainable fuel production. While CO₂ can be derived from a wide range of sources, CH₄ primarily originates from deep-sea sediments, oil and gas fields, and biological synthesis [1,2]. By transforming two major climate-forcing gases into clean chemical feedstocks, DRM offers a promising pathway for addressing global climate challenges [3–5]. However, its industrial application faces significant obstacles, including inefficient thermal management, catalyst deactivation caused by carbon deposition, and the strongly endothermic nature of the reaction ($\Delta H = +247$ kJ/mol) [6–8].

Owing to its high energy demand, DRM typically requires high operating temperatures in the range of 800–1000 °C to achieve desirable conversion rates. Conventional heating methods often result in uneven temperature distribution, promoting side reactions such as methane decomposition (CH₄ → C + 2H₂) and CO disproportionation (2CO → C

+ CO₂). These reactions contribute to carbon buildup on catalysts, particularly on widely used nickel-based materials [9–13]. The mechanisms underlying the catalyst deactivation have been explored in previous studies. For example, Kroll et al. [8] observed that nickel-based catalysts deactivate rapidly owing to the formation of filamentous carbon. Osaki et al. [10] found that the rate of carbon deposition is strongly influenced by metal-support interactions and local reaction kinetics. To explain these kinetics at the molecular level, Bradford and Vannice [14] highlighted the crucial role of CH_x intermediates and their interaction with surface oxygen derived from CO₂ dissociation in determining catalytic performance. A study of Ni/Al₂O₃ catalysts by Zhang and Verkyios [15] further demonstrated that the rate-determining step (RDS) shifts from CH₄ dissociation at lower temperatures to surface carbon oxidation at higher temperatures. Collectively, these studies highlight the crucial role of dynamic thermal control in sustaining catalyst performance and minimizing deactivation.

Recent advancements in catalytic materials have significantly improved DRM efficiency. For instance, Sun et al. [16] demonstrated

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that concerted catalysis by single atoms and nanoclusters enhances bioethanol activation and dehydrogenation, suggesting that synergistic effects among active sites could be harnessed to improve DRM reactivity. Such developments in catalyst design complement thermal management strategies, offering a comprehensive approach to optimizing DRM. Innovative reactor configurations have also been explored to improve heat transfer and reaction efficiency. Wang et al. [17] conducted a numerical study of DRM in a disk reactor using a focused solar simulator, achieving a thermochemical energy storage efficiency of 28.4 %, surpassing that of traditional tubular reactors. This highlights the potential of unconventional reactor geometries to improve DRM performance through improved thermal distribution. Microwave heating, an alternative that provides direct heating of polar molecules, has been widely applied in fields including chemical engineering [18–21]. Its application in DRM has also attracted extensive research interest due to its significant potential [22]. In refining reactor designs, computational fluid dynamics (CFD) plays a crucial role. Ghazanfari et al. [23] used CFD to analyze twisted tube configurations in helical heat exchangers, showing that optimized pitch lengths can increase thermal efficiency by up to 22 % at higher Reynolds numbers. These findings underscore the value of CFD in guiding reactor design for improved heat and mass transfer, which are key factors in DRM systems.

Electromagnetic induction heating (EIH), a technique that enables rapid localized heating via electromagnetic fields, is a promising solution for addressing thermal management challenges in the DRM process. Unlike conventional heating methods, EIH can deliver energy directly into the reaction zone by generating heat within conductive materials through induced eddy currents. This allows for rapid temperature increases (up to 100 K/min) and uniform heat distribution [24–26]. This enhanced thermal uniformity and fast response capability helps maintain efficient syngas production [27–29]. Moreover, EIH enables real-time modulation of heating by adjusting electromagnetic parameters, offering dynamic control that is unattainable with traditional furnaces. This makes EIH particularly well suited for managing the delicate kinetic balance of the DRM reaction [30–32]. The distinct heating effects produced by low- and high-frequency fields further demonstrate EIH's capability to regulate temperature in catalytic multiphase processes. This is particularly important for heating catalysts in fixed-bed reactors under unsteady flow conditions, where EIH improves process management, energy efficiency, reactor design simplification, and operational safety [33]. Xiong et al. [34] utilized induction heating to rapidly fabricate various electrocatalysts on a foam nickel substrate for electrochemical water splitting. The induction heating rate was fast (up to 10 °C/s in air), enabling the synthesis of catalysts with specific nanostructures and excellent properties within a short period of time. This rapid, simple, and effective synthesis method has broad application prospects in the fields of energy storage and conversion, reducing costs by shortening the preparation time and improving the performance of the catalysts. Yu et al. [35] proposed an EIH pyrolysis reactor and studied the effects of coil parameters on the temperature field and the production of undecylenic acid methyl ester and heptanal. Their study found that the simulated data closely matched experimental results, verifying the feasibility of multiphysics numerical models (electromagnetics, heat transfer, and reaction kinetics) for optimizing chemical reactions in EIH reactors.

EIH systems are inherently modular; induction coils can be scaled in terms of length and number of turns to match larger reactor volumes, as demonstrated in industrial-scale applications. This modularity allows flexible adaptation to different production capacities without sacrificing heating uniformity. Although EIH systems experience some energy leakage through electromagnetic waves, their ability to focus heating on metallic materials near the reaction zone can improve energy efficiency compared to conventional heating methods. Optimized designs can also yield more uniform temperature distribution, which can potentially reduce carbon deposition and extend the lifespan of the catalyst, as observed in DRM stability tests. EIH offers advantages in long-term

operational stability. Steel reactor walls exhibit resistance to cyclic induction heating, and the reduced carbon deposition resulting from uniform temperatures contributes to longer catalyst lifespans. In the context of green energy, EIH presents a scalable and sustainable solution for the large-scale transformation of chemical manufacturing toward environmentally responsible production.

Recent advancements in electromagnetic sensing technologies, such as the permanent-magnet-based flow velocimeter developed by Wang et al. [36], offer insights into optimizing fluid dynamics in reactors. Their work, which minimizes the output drift by measuring current instead of voltage, highlights the potential for precise flow monitoring in DRM reactors to enhance reactant distribution and reaction efficiency. Cao et al. [37] studied the pore size and material composition of porous foam within a metal tube, optimized the related electrical parameters, and provided important ideas for industrial applications.

Numerical simulations are important tools for investigating multiphase processes, interfacial phenomena, and optimizing EIH in DRM systems prior to industrial implementation. At the molecular level, advanced techniques such as molecular dynamics (MD) simulations are valuable for elucidating interfacial phenomena that affect catalyst performance and deactivation [38–42]. However, bridging the gap between nanoscale understanding and practical reactor-scale applications remains a significant challenge. In contrast, CFD is a well-established method for modeling chemical reactions in industrial systems. Widely applied in combustion, chemical synthesis, and pollution control, CFD modeling is crucial for optimizing DRM reactors by evaluating catalyst-flow interactions, as reviewed by Kaydoun et al. [43]. By predicting kinetics, transport phenomena, and thermal fields across the entire system, CFD enables robust chemical process design. For reactors driven by EIH, this framework is extended into a multiphysics model that integrates CFD with Maxwell's equations. Coupled simulations have proven that this approach is effective for optimizing the EIH process, showing that parameters such as coil geometry and current frequency significantly impact thermal uniformity and control response rates [30,44,45].

Despite significant progress in catalyst development and kinetic modeling, the integration of EIH into DRM remains insufficient, especially because of the scarcity of multiphysics simulation studies. This study aims to fill these gaps through a systematic approach: First, we develop a multiphysics coupling model using ANSYS-Maxwell and ANSYS-Fluent to simulate the EIH-assisted DRM reactor. This model quantitatively characterizes the “electricity–magnetism–electricity–heat” energy conversion pathway and maps the distribution characteristics of electromagnetic fields and heat transfer. Second, we design comparative scenarios involving both conventional reactors and EIH-coupled reactors to analyze differences in temperature distribution within the reaction zone, as well as conversion rates of CH₄ and CO₂ between EIH and traditional preheating methods. Finally, we reveal the mechanism by which EIH enhances DRM performance, providing a theoretical basis for optimizing coil parameters (such as frequency and structure) and reactor design. By detailing this integrated process, this study offers critical insights for developing cleaner, more efficient syngas production pathways, contributing to sustainable greenhouse gas utilization.

2. Reactor design

As shown in Fig. 1(a), the two-dimensional structure of the basic reactor consists of three sections. The first section is the inlet cavity area, and the third section is the outlet cavity area, both with a length of 50 mm. The middle section is a porous area with a length of 100 mm. The reactor has an internal diameter of 15 mm, a total length of 200 mm, and a wall thickness of 2 mm. The tube wall is made of steel. The porous area is the reaction zone, where the catalyst is stacked. A full three-dimensional view is shown in Fig. 1(b).

For the DRM reaction, a commonly used Ni-based catalyst is

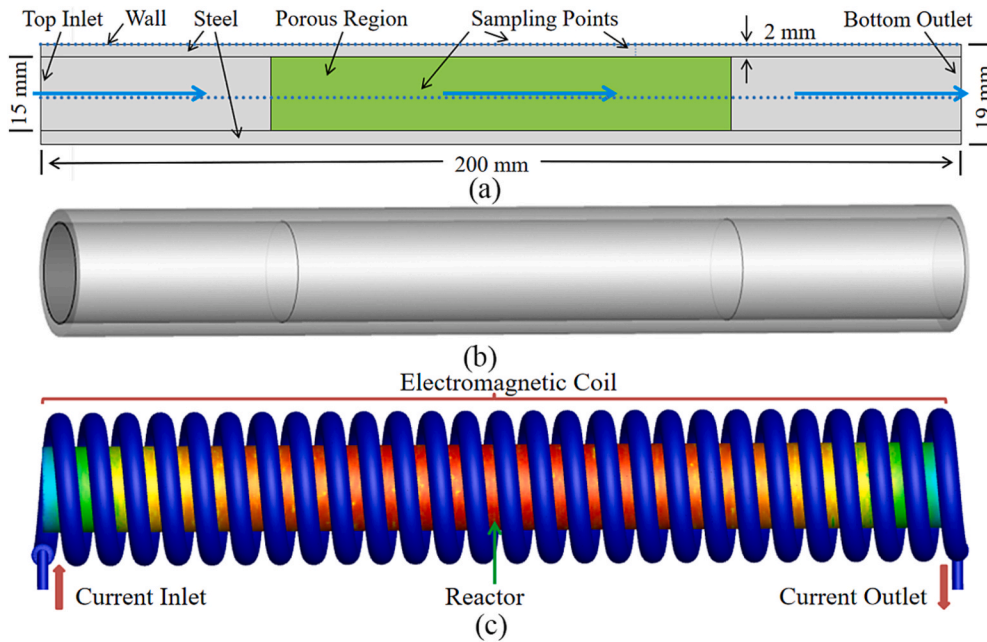


Fig. 1. Schematic diagrams of the DRM reactor: (a) Two-dimensional sectional view of the reactor; (b) three-dimensional diagrams of the reactor; (c) reforming reactor 3 coupled with electromagnetic coils.

employed. The reactants, CH_4 and CO_2 enter the inlet cavity area from the top inlet end at a specified ratio, velocity, and temperature, with relevant parameters shown in Table 1. The gas mixture flows into the porous reaction zone for catalytic conversion. The outer boundary of the steel wall is assumed to be adiabatic. Given that this reaction is endothermic, without energy supplementation, the temperature in the reaction zone will inevitably decrease. After the reaction, the mixture of reactants and products flows into the outlet cavity and exits the reactor from the bottom outlet. As temperature decreases, the reaction efficiency declines, and the process may weaken or even cease if the reactor is extended without energy supplementation.

The reactor with added EIH is shown in Fig. 1(c). An electromagnetic induction coil is wrapped around the reactor. The current enters through the current inlet, circulates through the electromagnetic induction coil, and exits via the current outlet. The relevant parameters are shown in Table 2.

To determine the differences between a reactor equipped with EIH and a conventional reactor, three working conditions were established: inlet temperatures of 1000 K, 1100 K, and 1000 K coupled with electromagnetic excitation, defined as Reactors 1, 2, and 3, respectively. Applying EIH to the whole Reactor 3 instead of only the reaction zone reduced magnetic field and heat losses typically caused by shorter coils. In addition, at a fixed heating rate, a longer EIH coil helped minimize axial temperature gradients and suppressed side reactions. In this study, a conventional heating method with a higher inlet temperature (Reactor 2) was used as a preliminary reference. While increasing the inlet temperature further could elevate the reaction zone temperature to match or

Table 2

Physical properties and operating conditions of electromagnetic coils for Reactor 3.

Solution type	Eddy current
Excitation current	20 A
Adaptive frequency	1 KHz
Coil material	Copper
Coil radius	2.5 mm
Helix radius	14.5 mm
Pitch	7.5 mm
Turns	27

exceed that of the EIH setup, it would also result in excessively high temperatures at the entrance of the reactor zone. This leads to severe side reactions and significantly reduces catalyst durability.

3. Numerical models and set-ups

The simulation procedure consisted of five key steps:

- (1) Model Development: Constructing multiphysics models incorporating electromagnetics, heat transfer, and reaction kinetics for methane dry reforming reactors.
- (2) Electromagnetic Simulation: Performing 3D numerical simulations to calculate the Joule heat source density (applicable only to EIH cases).
- (3) Coupled Equation Solving: Using the derived Joule heat source density (if available) to solve 3D coupled governing equations for hydrodynamics and chemical kinetics.
- (4) Performance Analysis: Evaluating CH_4/CO_2 conversion rates, temperature distribution, velocity fields, and concentration profiles.

This structured approach enabled a comprehensive evaluation of reactor performance from initial modeling to final validation.

According to the principle of electromagnetic induction in reactor 3, when an electric current is applied, an alternating magnetic field is generated around the induction coil, which induces an electromotive

Table 1

Physical properties and operating conditions of all reactors.

Reactor diameter	0.019 m
Temperature of inlet	reactor 1,3 (1000 K) reactor 2 (1100 K)
CH_4 , CO_2 , molar ratio of inlet	1:2
Pressure of inlet	10 bar
Porosity	0.3
Catalyst	nickel-based
GHSV	2000/h
Heat transfer coefficient	0
Diameter of catalyst particles	0.6 mm

force and eddy currents inside the conductor. This process can be described by the following electromagnetic equations [46]:

$$\nabla \times \vec{H} = \vec{J}, \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}, \nabla \cdot \vec{E} = \nabla \cdot \vec{B} = 0, \quad (1)$$

$$\vec{B} = \mu \vec{H}, \vec{J} = \sigma \vec{E}, Q = \vec{J} \cdot \vec{E}, \delta = \sqrt{\frac{1}{\pi \mu f \sigma}}. \quad (2)$$

Here, $\vec{B}$ is the magnetic induction intensity, $\vec{H}$ is the magnetic field intensity, $\vec{E}$ is the electric field intensity, μ is the relative magnetic permeability of the conductor, $\vec{J}$ is the current density, σ is the electrical conductivity, δ is the skin depth, and f is the coil excitation frequency. During the induction heating process, the alternating electromagnetic field generates an eddy current density $\vec{J}$. A heat source density Q is generated when the eddy current flows within the steel wall, which is described by Joule's law.

For all reactors, the hydrodynamic model consists of continuity, momentum, species, and energy equations [47–49]:

$$\nabla \cdot (\rho \vec{v}) = 0 \quad (3)$$

$$\nabla \cdot (\rho \vec{v} \vec{v}) = -\nabla p + \nabla \cdot \vec{\tau} + \rho \vec{g} + S_i \quad (4)$$

$$\nabla \cdot (c_i \vec{v}) = \dot{q}_i + \nabla \cdot \vec{Y}_i \quad (5)$$

$$\nabla \cdot [\vec{v} (\rho E_f + p)] = H + \nabla \cdot \left[(\epsilon k_f + (1 - \epsilon) k_s) \nabla T + \vec{\tau} \cdot \vec{v} + \sum_i h_i \vec{Y}_i \right] \quad (6)$$

with the drag term following the Darcy–Forchheimer model [50,51] and the Ergun equation[52].

$$\vec{S} = -\left(\frac{\mu}{\beta} \vec{v} + \frac{1}{2} C \rho |\vec{v}| \vec{v}\right), \beta = \frac{D^2 \epsilon^3}{150(1 - \epsilon)^2}, C = \frac{3.5(1 - \epsilon)^2}{D \epsilon^3} \quad (7)$$

Here, ρ is the density, p represents the pressure, $\vec{v}$ is the apparent velocity, μ is the hydrodynamic viscosity, D is the particle diameter, ϵ is the porosity, β is the permeability, C is the inertial resistance factor, c_i is the volumetric concentration of component i , E_f is the total energy of the fluid, H is the enthalpy source term (from reaction), T is the temperature, k_f/k_s are the fluid/solid thermal conductivities, $\dot{q}_i$, h_i , and $\vec{Y}_i$ are the source (from reaction), enthalpy, and diffusion flux of component i , respectively. The stress tensor and concentration flux vectors should be specified using turbulence models.

The reaction kinetics follows the Arrhenius equation:

$$r_R = k_R C_{CH_4} C_{CO_2}, k_R = A \exp\left(-\frac{E}{RT}\right). \quad (8)$$

Here, r_R is the reaction rate, C_{CH_4} and C_{CO_2} are the component concentrations, the gas constant $R = 8.413 \text{ J/(mol} \cdot \text{K)}$, the pre-exponential factor $A = 2.42 \times 10^9 \text{ mol/g}_{\text{cat}} \cdot \text{s} \cdot \text{Pa}^2$, and the reaction activation energy $E = 2.37 \times 10^4 \text{ J/mol}$ [53]. This power-law kinetics is applicable under limited conditions but is appropriate for the present comparative study of EIH.

Simulations of the EIH process and DRM reaction, involving electromagnetic equations, hydrodynamic models, and kinetic models, were performed. Electromagnetic fields were solved using finite element discretization in ANSYS-Maxwell, while hydrodynamics were solved using the SIMPLE algorithm and the standard k - ϵ turbulence model in ANSYS-Fluent 2022. All cases were calculated until reaching steady state, with convergence determined by a residual threshold of 1×10^{-6} . Relevant data were analyzed and compared to assess the impact of EIH technology on the DRM reaction. The conversion rate of reactant i was

computed using the following formula:

$$\varphi_i = \frac{\int_{\text{inlet}} c_i v ds - \int_{\text{outlet}} c_i v ds}{\int_{\text{inlet}} c_i v ds} \times 100\% \quad (9)$$

As sketched in Fig. 2, an adaptive structured mesh was adopted. The reactor was meshed with approximately 740,000 hexahedral elements, while the coil was meshed with approximately 420,000 tetrahedral elements. For the mesh convergence study, simulations were also performed using finer meshes: 1,090,000 elements for the reactor and 450,000 for the coil. By comparison, it was found that the conversion rates were highly consistent: $\varphi_{CH_4} = 10.81\%$ and $\varphi_{CO_2} = 14.84\%$ from the sparser mesh, and $\varphi_{CH_4} = 10.93\%$ and $\varphi_{CO_2} = 14.96\%$, with relative deviation being approximately 1%. This indicated that the sparser mesh was sufficient for accurate results. To improve computational efficiency, the model with fewer grids was selected for calculation. The shapes and distributions of grids are sketched (with reduced density) in Fig. 2.

4. Results and discussion

4.1. Electromagnetic fields

A numerical simulation of the EIH for the DRM reactor was conducted, and the following electromagnetic-induction-related data were obtained and analyzed.

Fig. 3 shows the contours of the distribution of magnetic induction intensity amplitude B in the reactor. The original source of the alternating magnetic field was the alternating current in the coil, and the current induced by the alternating magnetic field inside the conductor will in turn weaken the magnetic field, causing the skin effect of the magnetic field. Fig. 3(a) shows the distribution of the magnetic induction intensity on the outer wall of Reactor 3. It can be clearly seen that the magnetic field in the middle region is strong, with the B value reaching 1.46 T, while the magnetic field at both ends is weak, with a B value of approximately 0.73 T, which corresponds to the concentration of the magnetic field excited by the coil in the middle of the reactor. Figs. 3(b), (c), and (d) show the contours of the magnetic induction intensity in the cross sections of the top, middle, and bottom parts, respectively. The B values at the top and bottom are close to 0, while the middle part is distributed annularly along the wall, which is consistent with the skin effect (high-frequency magnetic fields are concentrated on the conductor surface, and the internal magnetic field is weak).

In addition, Fig. 4 shows the contour of the magnetic field intensity amplitude distribution H in the reactor. Fig. 4(a) shows the distribution of the magnetic field intensity on the outer wall of Reactor 3. It can be clearly seen that the magnetic field in the middle is strong, with the H value reaching 1200 A/m, and the magnetic field at both ends is weak, with the H value essentially below 600 A/m, which is consistent with the distribution trend of the magnetic induction intensity. However, because magnetic permeability is related to temperature, H is more concentrated in the middle of the reactor than B . Figs. 4(b), (c), and (d) show the contours of the magnetic field intensity in the cross sections of the top, middle, and bottom parts, respectively. The H values at the top

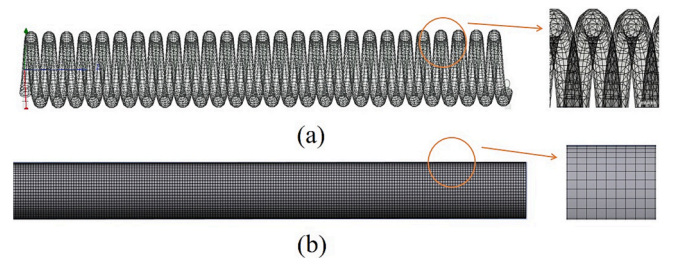


Fig. 2. Coarsened sketches of mesh of the DRM reactor: (a) the reactor; (b) the coil.

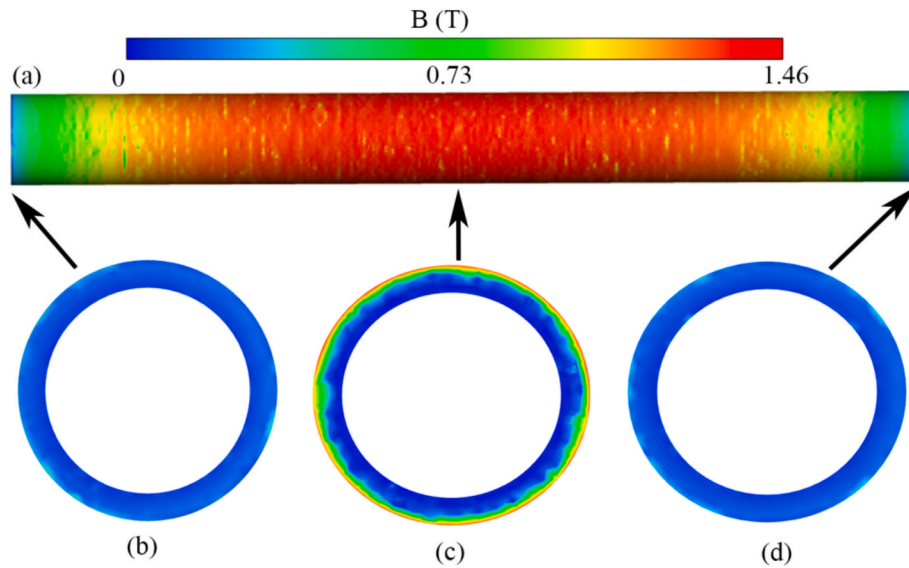


Fig. 3. Distribution contours of the magnetic induction intensity of Reactor 3: (a) at the wall surface; (b) at the cross-sections of the top, middle, and bottom.

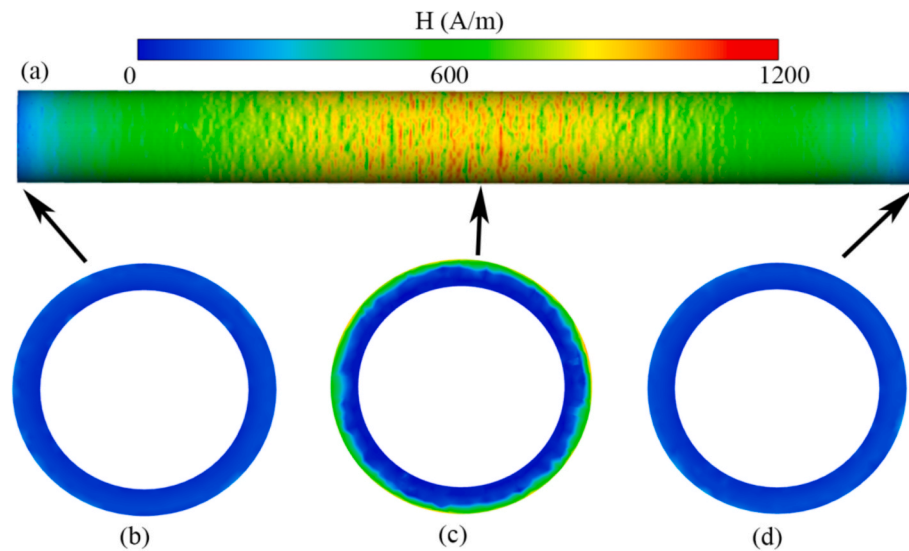


Fig. 4. Distribution contours of the magnetic field intensity of Reactor 3: (a) at the wall surface; (b) at the cross-sections of the top, middle, and bottom.

and bottom tended toward zero, while the middle part was distributed annularly along the wall, matching the annular distribution of B .

Furthermore, Fig. 5 shows the contour of the electric field intensity amplitude E distribution in the reactor. An electric field is the induced electric field induced by an alternating magnetic field, as shown in Fig. 3. In the EIH, the induced electric field drives the movement of free electrons in the conductor, generating an induced current (eddy current) and heat through the Joule effect. Fig. 5(a) shows the distribution diagram of the electric field intensity on the outer wall of Reactor 3. The electric field in the middle is strong, with the E value reaching 1.2 V/m, and the electric field at both ends is weak, with the E value essentially below 0.6 V/m. Figs. 5(b), (c), and (d) show the contours of the electric field intensity in the cross sections of the top, middle, and bottom parts, respectively. The electric field intensity at the top and bottom is very weak, while the middle part is distributed annularly along the wall. In contrast to the strong skin effect of B , the distribution of E was not highly concentrated on the outer wall of the reactor, resulting in the distribution of the induced current and corresponding Joule heat source being closer to the reaction zone.

Fig. 6 shows the contours of the current density distribution within the reactor, which is driven by the induced electric field. The current density distribution directly determines the power density of Joule heating and serves as a direct measure of the heating effect as indicated by Eq. (2). Fig. 6(a) shows the current density distribution diagram along the outer wall of Reactor 3. The current in the middle is strong, with the J value reaching $2.4E + 6 \text{ A/m}^2$, which is consistent with the high-value region of the electric field intensity, reflecting that the induced electric field drives free electrons to form eddy currents. Figs. 6(b), (c), and (d) show the contours of the current density in the cross sections of the top, middle, and bottom parts, respectively, which are completely synchronized with the annular distribution of the electric field intensity, conforming to the eddy current formation mechanism. This distribution supports the design choice of wrapping the entire reactor with a coil. The Joule heat generated by the current density is mainly concentrated in the central region, approximately half the length of the reactor, which corresponds to the region filled with the catalysts.

To demonstrate the distribution of multiple physical fields quantitatively, Fig. 7 shows the variation in field amplitudes along the height

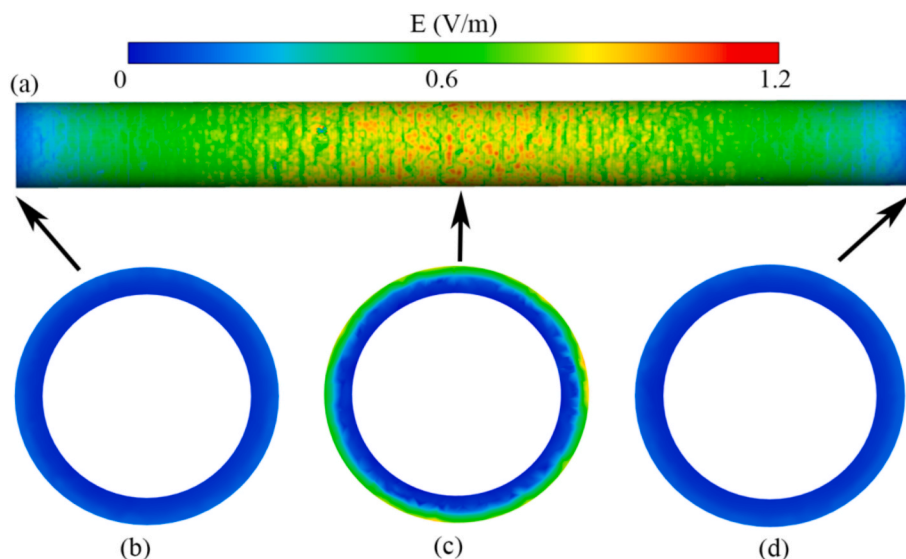


Fig. 5. Distribution contours of the electric field intensity of Reactor 3: (a) at the wall surface; (b) at the cross-sections of the top, middle, and bottom.

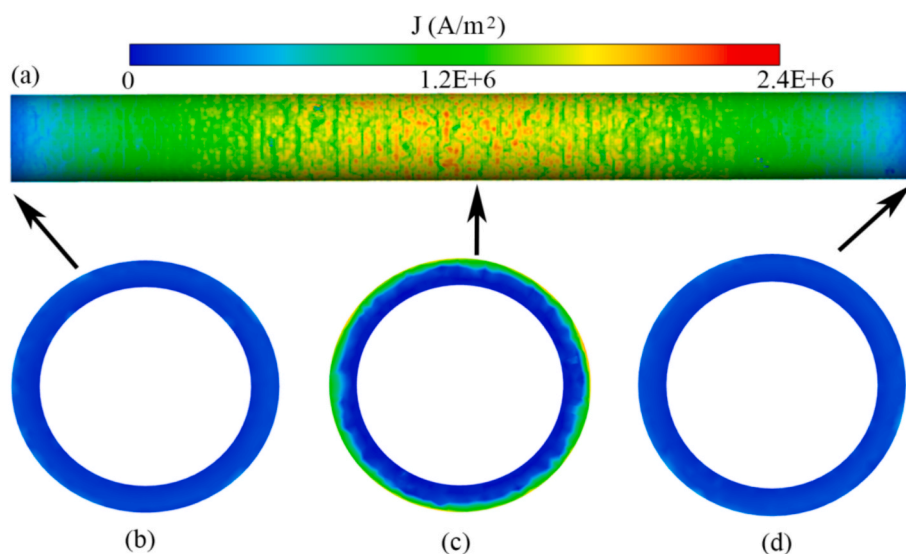


Fig. 6. Distribution contours of the current density of Reactor 3: (a) at the wall surface; (b) at the cross-sections of the top, middle, and bottom.

of the reactor's outer wall, and Fig. 8 presents the radial distribution of field amplitudes in the middle cross-section of the reactor. In Fig. 7, all fields (B , H , E , J) are strongest in the middle and weakest at both ends. This indicates that the effects of the magnetic field, electric field, and eddy current induced by the coil are concentrated in the middle region of the coil (accounting for approximately 50 % of the coil length). This design aligns with the fact that the reaction zone occupies 50 % of the reactor's length, demonstrating a deliberate design strategy to target heating where it is most needed for the endothermic DRM reaction. Fig. 8 shows that field intensity is concentrated near the outer wall, with the outer wall being much stronger than the inner wall. This embodies and verifies the skin effect of high-frequency time-varying magnetic fields and reflects that the Joule heat source is more concentrated on the outer wall of the reactor. Meanwhile, heat generated in the middle of the reactor can be more effectively transferred to the reaction zone. If required in other specific application scenarios, the heating depth can be increased by reducing the coil current frequency.

In summary, the axial distributions of all fields (B , E , H , and J) in the figures are highly consistent, with concentrations centered around the middle wall of the reactor. Behind this lies the core physical mechanism

of “electricity–magnetism–electricity–heat” in EIH: a time-varying magnetic field applied by an external coil induces a vortex electric field. The eddy current driven by the vortex electric field, on one hand, offsets part of the magnetic field inside the conductor, and on the other hand, generates Joule heat. These field distributions can intuitively demonstrate the effect of EIH, providing a key basis for the optimal design of reactors. Specifically, the field distribution can be optimized to improve heating efficiency and uniformity by adjusting the coil current frequency (to control the skin depth), coil structure (to optimize the magnetic field distribution), and material (to adjust the electrical/magnetic permeability). For example, low frequencies are suitable for deep heating of thick walls, while high frequencies enhance surface heating; short coils optimize heating targeting, and long coils improve heating uniformity.

4.2. Temperature fields

The effect of EIH must be analyzed more specifically in combination with the reactor's temperature distribution and reaction efficiency. Fig. 9 shows the axial cross-sectional temperature field contours of the

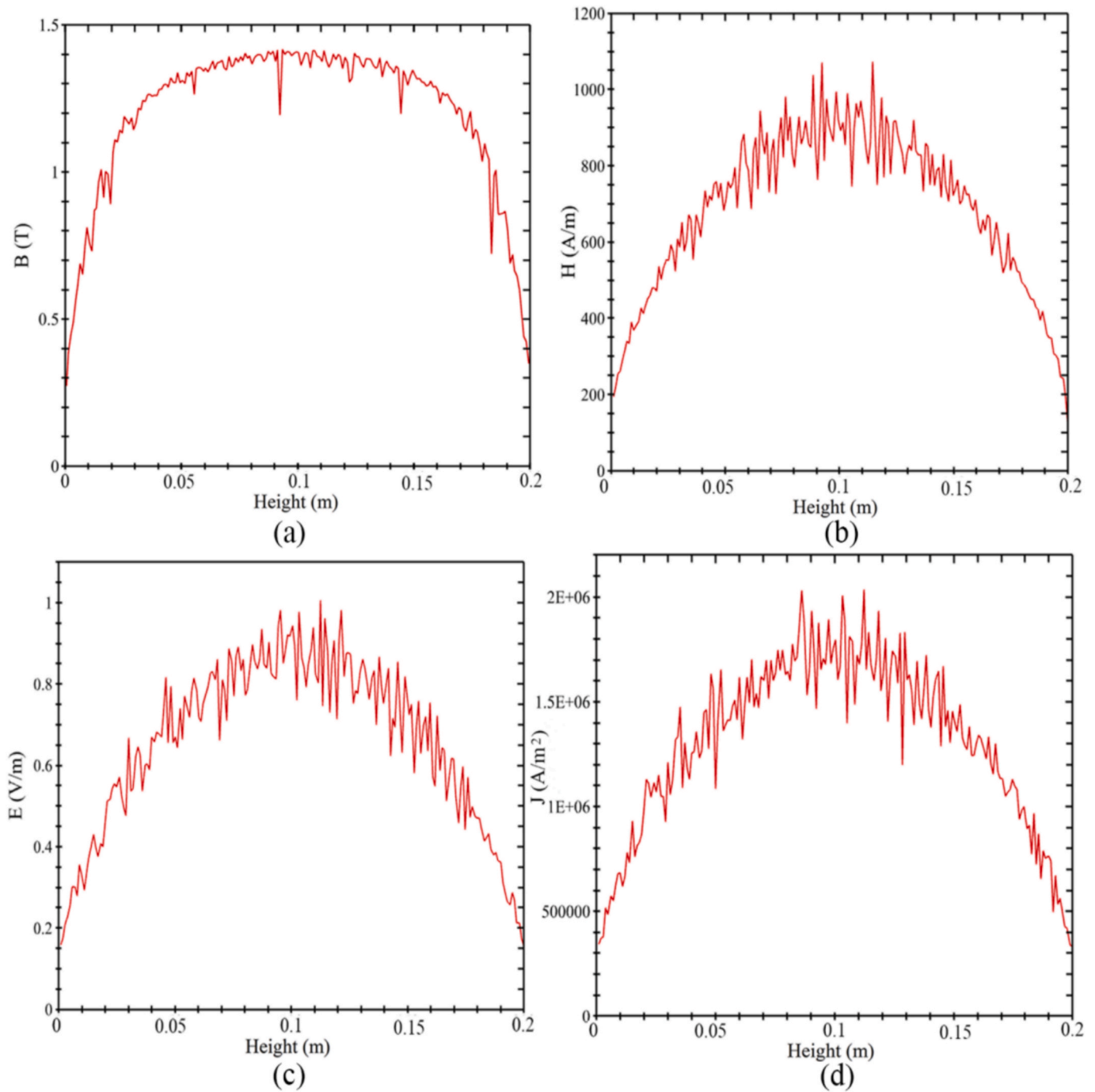


Fig. 7. Curves of the sampling positions of the wall surface along the direction of height: (a) magnetic induction intensity; (b) electric field intensity; (c) magnetic field intensity; (d) current density.

three reactors and highlights the transformative impact of EIH on thermal management in DRM. As shown in Figs. 9(a) and 9(b), without EIH, the temperature contour is dominated by heat loss from the endothermic reaction. In the absence of an external magnetic field, the temperature in the inlet cavity area is significantly lower than in other regions and continuously decreases sequentially from the middle reaction zone to the outlet cavity area. More importantly, the low temperature in the reaction zone is conducted upstream through the steel wall, resulting in a rapid temperature drop near the inlet. This is due to the cooler gas in the middle and rear sections lowering the tube wall temperature, which in turn reduces the temperature in the inlet cavity area via solid heat conduction.

The temperature field in Fig. 9(c) shows a markedly different trend. Owing to the effect of EIH, the temperature field distribution is reversed, the tube wall temperature is consistently higher than the fluid temperature, and the adjacent fluid is heated through heat conduction. Therefore, the temperature of the entire reactor increases continuously along the flow direction. Notably, because the endothermic effect occurs in the reforming reaction in the porous reaction zone, the temperature rise in the first half of the reaction zone along the flow direction is not obvious. As the reaction rate in the second half of the reaction zone decreases with the consumption of reactants, the endothermic effect of the reaction gradually weakens; thus, the temperature in the second half of the reaction zone increases significantly, and after the fluid flows out

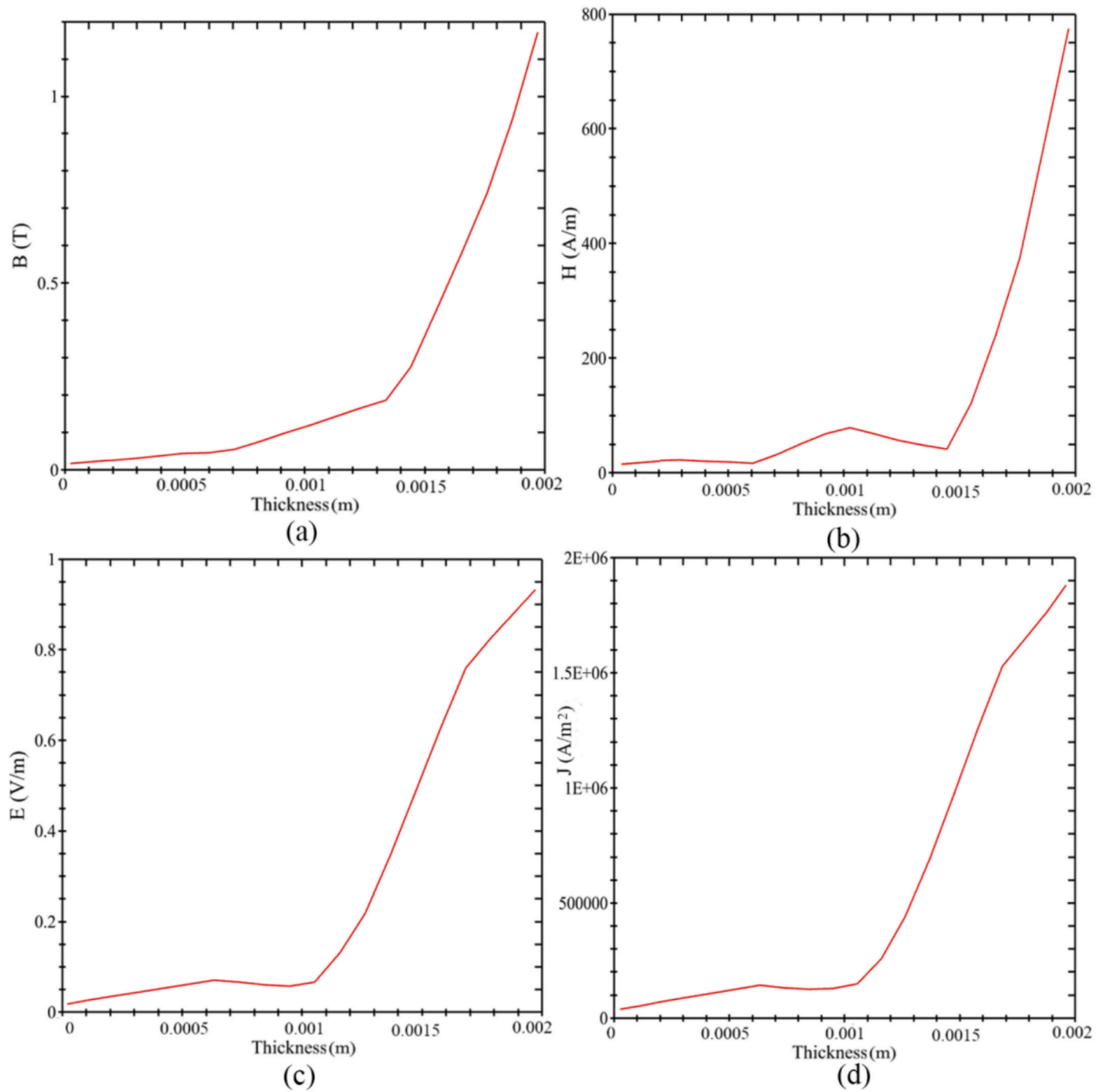


Fig. 8. Curves of the sampling positions of the middle cross-section along the wall thickness: (a) magnetic induction intensity; (b) electric field intensity; (c) magnetic field intensity; (d) current density.

of the reaction zone, the heating effect becomes even more pronounced.

To supplement the contours, Fig. 10 shows the temperature distribution along the central axis of the reaction zone. Reactors 1 and 2 exhibit the same temperature change trend; the temperature decreases gradually in the outlet cavity area, drops sharply at the front end of the reaction zone, decreases slowly in the middle and rear sections of the reaction zone, and remains essentially unchanged after entering the outlet cavity area. The temperature change trend along the central axis of Reactor 3 is almost opposite to that of Reactors 1 and 2, increasing continuously along the flow direction.

Notably, the reaction rate only depends on the temperature within the reaction zone, which is also the effective temperature zone for the reaction. In the reaction mechanism adopted in this study, the higher the

effective temperature zone, the higher the reaction rate. Even though the inlet temperature of Reactor 3 is lower than that of Reactor 2, after EIH, the temperatures of most of the reaction zones in Reactor 3 were higher than those in Reactor 2, indicating that EIH can efficiently promote the conversion rate. In addition, the temperature in the outlet cavity area of Reactor 3 continued to increase. Therefore, in the common production scenario where multiple reactors are used in series, the additionally heated fluid in the outlet cavity area can provide energy for subsequent endothermic reactions.

4.3. Conversion rates

To quantify the efficiency of the DRM reactions in the three reactors,

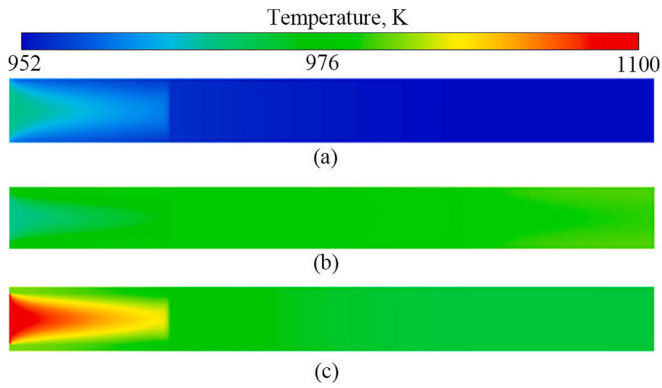


Fig. 9. Contours of the cross-section temperature field distribution: (a) Reactor 1; (b) Reactor 2; (c) Reactor 3.

the conversion rates of CH_4 and CO_2 in the three reactors were calculated. The specific data are shown in Table 3, which confirms the superiority of EIH in optimizing DRM efficiency. The conversion rates follow the given order from highest to lowest: Reactor 3 > Reactor 2 > Reactor 1. Although the inlet temperature of Reactor 2 reached 1100 K, which was higher than that of Reactor 3 (with an inlet temperature of 1000 K coupled with electromagnetic excitation), the conversion rate was still lower than that of Reactor 3. This is because, under EIH, the temperature of the effective temperature zone in Reactor 3 is significantly higher than that in Reactor 2 (see Fig. 10). Furthermore, it can inhibit the side reactions that are prone to occur in an environment with

uneven temperatures.

4.4. Concentration fields

Based on the analysis of the various data in the previous text, a further study was conducted on the concentration fields of both reactants and products.

Fig. 11 presents the contours of the CO_2 concentration field distribution on the axial cross-sections of the three reactors, all of which intuitively display obvious concentration gradients—clear evidence that the DRM reaction is primarily occurring within the porous zone. These gradients arose from the consumption of CO_2 as the reaction proceeded, with steeper gradients indicating more intense local reaction activity.

Focusing on the outlet CO_2 concentrations, Figs. 11(a) and 10(b) compare reactor 1 (1000 K inlet, no induction heating) and reactor 2 (1100 K inlet, no induction heating). Reactor 1 exhibited an outlet CO_2 concentration of 58.7 %, while Reactor 2 showed a lower value of 57.4 %. This difference confirms that increasing the inlet temperature within the tested range (1000–1100 K) promotes DRM reaction progress: higher initial temperatures provide more thermal energy to drive the endothermic DRM reaction, leading to more complete CO_2 consumption and thus a higher conversion rate in Reactor 2 compared to that in Reactor 1. However, the improvement in Reactor 2 is limited by the gradual temperature drop in the reaction zone (as observed in Fig. 10), which slows down the reaction rate as the reactants move through the porous zone, preventing further significant reduction in CO_2 concentration.

In contrast, Fig. 11(c) shows that Reactor 3 (1000 K inlet with induction heating) achieved the lowest outlet CO_2 concentration of 56.8

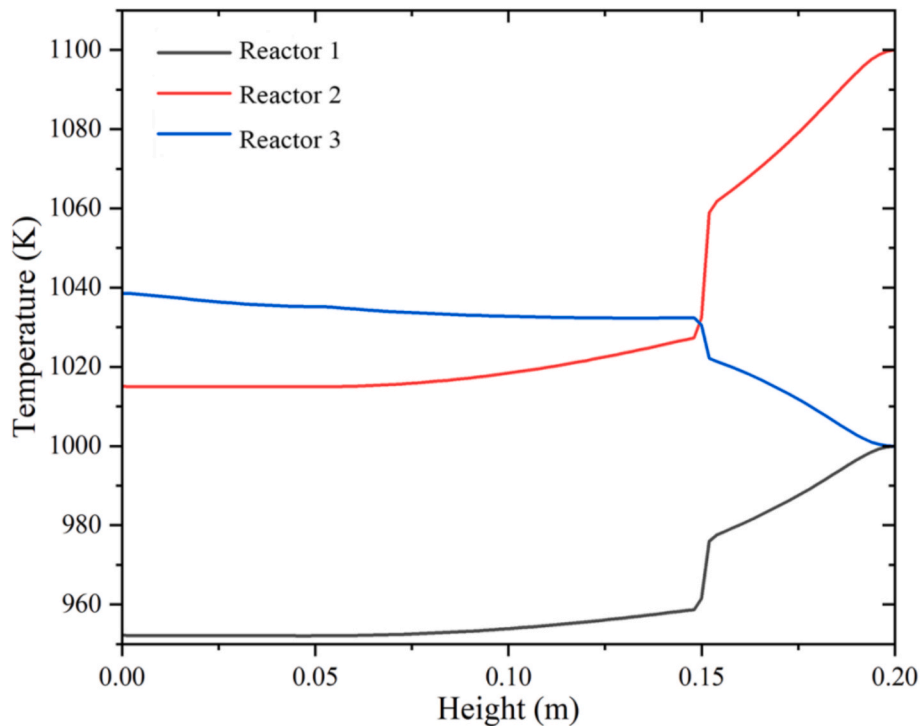


Fig. 10. Temperature curves of the sampling positions at the center of the reactor.

Table 3

Conversion rates of various DRM reactors.

Reactor	Electromagnetic excitation	Inlet temperature	CH_4 conversion rate	CO_2 conversion rate
1	×	1000 K	8.71 %	11.99 %
2	×	1100 K	10.21 %	13.94 %
3	✓	1000 K	10.81 %	14.84 %

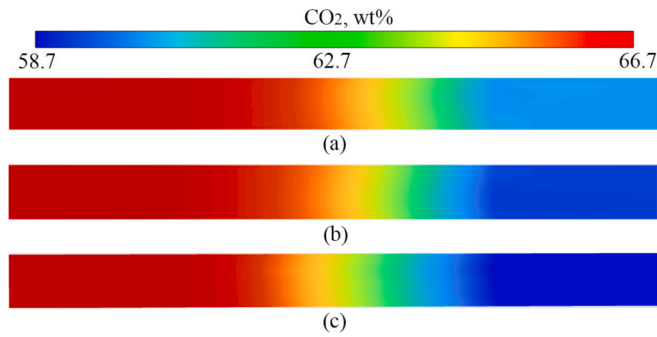


Fig. 11. Contours of the CO_2 concentration field distribution: (a) Reactor 1; (b) Reactor 2; (c) Reactor 3.

%. This superior performance, despite a lower initial inlet temperature than that of Reactor 2, stems from the sustained high temperatures in the reaction zone enabled by EIH. Unlike the passive temperature decline in Reactors 1 and 2, induction heating actively maintains the thermal

conditions where they are most needed (within the porous reaction zone), accelerating CO_2 consumption throughout the entire reaction process. Additionally, the concentration gradient in Reactor 3's porous zone is steeper than in the other two reactors, directly reflecting that induction heating enhances local reaction rates in the critical region, further driving down CO_2 concentration.

Moreover, Fig. 11 reveals that the concentration fields of all components are relatively uniformly distributed across the cross-sections, validating the use of central axis concentration curves for the quantitative analysis of reaction dynamics. This is further supported by Fig. 12, which plots the central axis concentration profiles of reactants (CH_4 , CO_2) and products (CO , H_2) for the three reactors, clearly illustrating the coupling between temperature fields (Fig. 10) and reaction kinetics.

For the reactants CH_4 and CO_2 , Reactor 3 exhibits the fastest consumption rate within the porous zone, with their concentrations dropping more steeply along the axis compared to Reactors 1 and 2. Conversely, the products CO and H_2 show the most rapid concentration increases in Reactor 3, peaking at the outlet of the porous zone. This trend aligns perfectly with the higher conversion rates reported in Table 3, confirming that induction heating not only maintains elevated

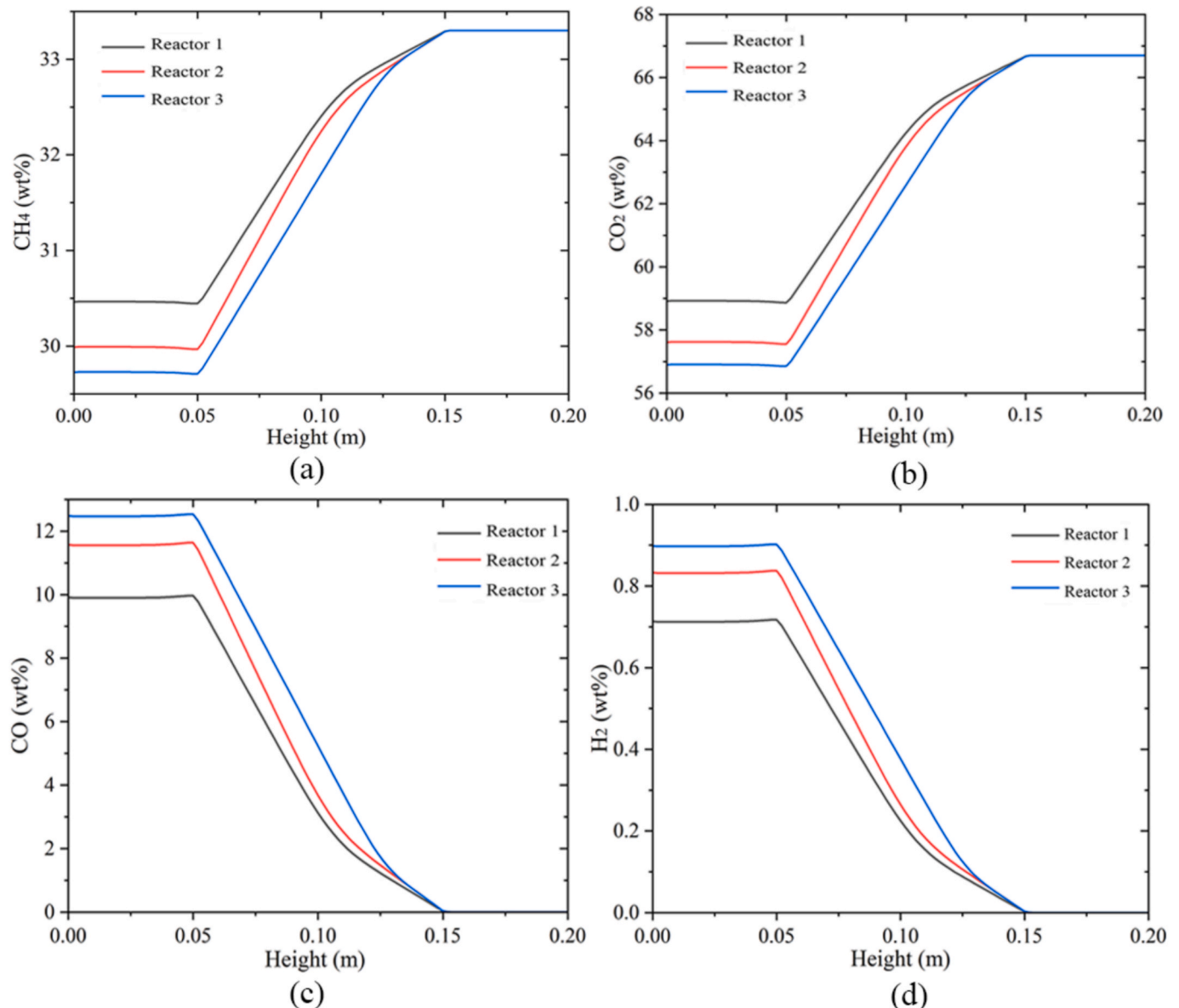


Fig. 12. CH_4 , CO_2 , CO , and H_2 concentration curves of the sampling positions at the center of Reactors 1, 2, and 3.

temperatures but also promotes more uniform and efficient utilization of reactants across the entire reaction zone.

Notably, in Reactors 1 and 2, the rate of concentration change slows significantly in the latter part of the porous zone. This indicates “reaction quenching,” a decline in reaction activity caused by temperature loss in these regions, as the initial thermal energy is gradually dissipated without external replenishment. However, in Reactor 3, this quenching effect was effectively mitigated by the continuous thermal input from induction heating, allowing the reaction to proceed vigorously throughout the porous zone.

In summary, Figs. 11 and 12 collectively demonstrate that induction heating is a more effective strategy than simply increasing the inlet temperature to enhance the DRM reaction performance. It achieves a higher reactant conversion, more complete product formation, and sustained reaction activity by maintaining optimal thermal conditions in the critical porous reaction zone.

4.5. Velocity fields

Since the space velocity was consistent, the velocity fields under the three conditions were essentially identical. Therefore, Reactor 1 was selected for further investigation. The velocity data of the axial cross-section of the reactor were examined. As shown in Fig. 13(a), the fluid velocity distribution contour map in the reactor shows a boundary layer effect in the cavity area; the flow velocity in the central area is significantly higher than that in the area near the tube wall. This boundary layer adheres to the classic Poiseuille flow characteristics, in which viscous forces near the wall slow down the fluid, creating a parabolic velocity profile with the highest speed along the axis. In the porous reaction zone, the flow becomes more uniform in the cross-section owing to the resistance of the porous medium, resulting in a lower peak flow velocity than that in the cavity area under a constant mass flow rate. The catalyst particles disrupted the laminar flow structure, dissipated kinetic energy, and flattened the velocity gradient across the section.

Fig. 14 shows the velocity curve along the central axis of the reactor, which also represents the peak velocity of each cross section. In the inlet cavity area, since the velocity distribution changes from a uniform distribution at the inlet to the appearance of a boundary layer, the peak flow velocity increases rapidly and maintains at a relatively high value; near the inlet of the porous zone, since the velocity tends to be uniform again in the cross-section, the peak flow velocity quickly drops back to the value at the inlet; in the porous reaction zone, owing to the occurrence of the catalytic reaction, the gas volume increases and density

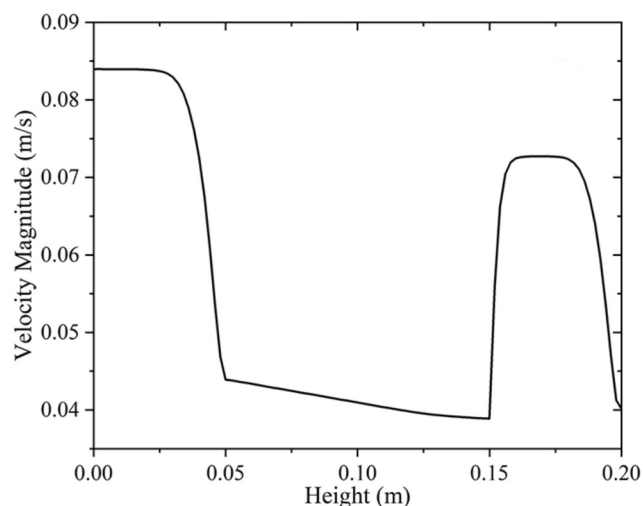


Fig. 14. Velocity magnitude curves of the sampling positions at the center of Reactor 1.

decreases; thus, the flow velocity increases under the constraint of constant mass flow rate, reducing density while preserving mass flow; after flowing out of the reaction zone, the flow velocity returns to the distribution with a boundary layer again, and the peak velocity increases rapidly and then remains unchanged. Fig. 13(b) shows a velocity-vector contour map, where five key parts are selected for magnification. The vector map clearly shows the changes in the fluid velocity magnitude and flow direction, reflecting the formation process of the boundary layer near the flow inlet and when flowing out of the porous zone as well as the velocity homogenization process when the flow enters the porous zone. These vector details help verify that flow uniformity in the reaction zone is maintained across all reactors, ensuring that thermal effects are the only variable affecting DRM performance, thus providing a critical phenomenological understanding for the study of DRM technology.

5. Conclusions

This study presents a comprehensive numerical investigation of methane dry reforming within an EIH reactor, integrating electromagnetics, heat transfer, fluid dynamics, and chemical kinetics. The

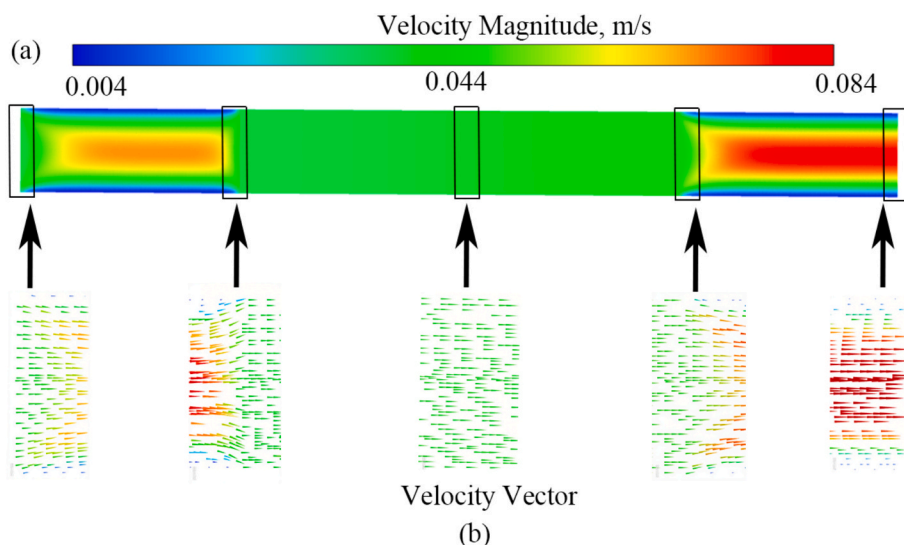


Fig. 13. (a) Contours of the velocity magnitude in the axial cross-section; (b) velocity vector distribution in transitional regions.

simulations reveal that EIH establishes an efficient “electricity–magnetism–electricity–heat” energy conversion pathway, in which a time-varying magnetic field induces a vortex electric field that drives eddy currents, generating localized Joule heating. The electromagnetic fields exhibit axial concentration, with peak intensity in the middle section of the reactor, and a radial skin effect, concentrating energy along the outer wall where the catalyst bed is located. This targeted heating approach ensures that thermal energy is effectively supplied to the most critical reaction zone.

Temperature analysis demonstrates that conventional reactors experience significant cooling along the flow direction due to the endothermic nature of the reaction, leading to reduced reaction rates and limited conversion. In contrast, the EIH-assisted reactor maintains a higher and more uniform temperature throughout the reaction zone. Remarkably, despite a lower inlet temperature compared to a conventional reactor, the EIH reactor achieves superior CH₄ and CO₂ conversion, highlighting the enhanced thermal management and reaction efficiency enabled by electromagnetic induction.

The concentration fields further confirm that EIH accelerates the consumption of reactants while promoting the formation of H₂ and CO. The combination of uniform heat distribution and elevated temperatures effectively suppresses side reactions and mitigates reaction quenching, ensuring sustained activity across the entire porous zone. The study also shows that heating performance can be tuned by optimizing coil design, adjusting operating frequency to control heating depth, and selecting appropriate material properties.

Overall, this study demonstrates that EIH is a highly effective strategy for improving the efficiency, uniformity, and sustainability of methane dry reforming. These findings provide valuable guidance for the design and optimization of industrial-scale reactors and offer a promising pathway toward cleaner, low-carbon syngas production. In addition, some findings in this study may provide valuable references for addressing the heat generation issue in electrical engineering.

CRediT authorship contribution statement

Zhenze Zhu: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Shengqi Zhang:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis. **Lifeng Xiao:** Supervision, Resources, Project administration, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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